

# NUWAN MADUSANKA

Full Stack Developer | Software Engineer | React · Next.js · Node.js · TypeScript

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## PROFESSIONAL SUMMARY

Results-driven Full Stack Developer with 3+ years of end-to-end software development experience across the complete web application lifecycle. Specializes in engineering high-performance, scalable SPAs and SSR solutions with React.js and Next.js, backed by robust Node.js/Express.js RESTful APIs and MongoDB data architecture. Proven track record of translating complex business requirements into clean, maintainable, production-grade code. Adept at Agile/Scrum delivery, CI/CD automation, and cross-functional collaboration — consistently delivering features on time and to specification in fast-paced enterprise environments.

## TECHNICAL SKILLS

**Frontend:** React.js, Next.js, JavaScript ES6+, TypeScript, HTML5, CSS3, Tailwind CSS, Bootstrap, Redux, Context API, Material-UI (MUI), CoreUI, Svelte, Nuxt.js, Vite, Radix UI, Framer Motion

**Backend:** Node.js, Express.js, RESTful API Design, JWT / OAuth2 Auth, ASP.NET Web API, Socket.IO, Stripe, Zod, Nginx, Jest, Vitest, Keycloak

**Database:** MongoDB, Mongoose, MS SQL Server

**DevOps & Tools:** Git, GitHub, Docker, CI/CD Pipelines, Azure DevOps, Huawei Cloud, Postman, Webpack, npm, Babel, Turborepo, pnpm, Cloudflare R2

**Core Strengths:** System Design, Agile/Scrum, Code Review, API Design, Performance Optimization, Cross-functional Collaboration, Problem Solving

**Languages:** Sinhala (Native), English (Professional)

## WORK EXPERIENCE

### DMS Software Engineering (Pvt) Ltd

Apr 2025 – Present

#### Software Engineer · Full-time · On-site

- Currently engineering DocFlo, an intelligent business process management platform built for modern enterprises to streamline operations, automate workflows, and boost productivity; contributing full-stack features across the React.js/Next.js frontend and Node.js/Express.js backend.
- Architect and ship end-to-end full-stack features using React.js, Next.js, Express.js, and Node.js, optimizing for performance, scalability, and SEO via server-side rendering.
- Lead system design sessions and technical requirement analysis to align engineering deliverables with business objectives, reducing rework by up to 30% by clarifying scope upfront.
- Champion code quality through rigorous peer code reviews, modular component architecture, and adherence to clean-code principles across the entire codebase.
- Streamline CI/CD pipelines and Agile sprint ceremonies, accelerating team delivery velocity by 20% and reducing deployment friction.
- Integrate third-party RESTful and microservice-based APIs, implementing JWT authentication and Keycloak-based role-based access control (RBAC) for secure data flows.

### DMS Software Engineering (Pvt) Ltd

Jun 2023 – Apr 2025

#### Associate Software Engineer · Full-time · Colombo, Sri Lanka

- Currently working on Docubinet — a robust, intelligent enterprise document content management system built on the MERN stack (MongoDB, Express, React, Node.js).
- Designed and implemented an intuitive Material-UI (MUI) frontend with Redux for state management, improving user experience and workflow efficiency.
- Integrated key functionalities such as QR scanning and digital signatures, significantly increasing security and usability within the system.
- Built a reusable, component-driven UI library with React.js and Tailwind CSS, cutting front-end development time by 35% across projects.
- Designed and implemented RESTful API endpoints with Express.js, enforcing secure authentication via JWT and data validation middleware.
- Contributed to CI/CD automation (Azure DevOps) and Agile sprint planning, improving sprint predictability by 25% and reducing release cycle time.
- Collaborated with cross-functional teams including UI/UX designers and Business Analysts; participated in client meetings to gather requirements and deliver effective solutions.

## SmartCode Software Solutions (Pvt) Ltd

May 2022 – Jun 2023

### Trainee Software Engineer · Full-time

- Spearheaded front-end architecture for a full-scale ERP system in the apparel manufacturing sector using React, HTML, CSS, and JavaScript, improving operational workflow efficiency for end users.
- Developed server-side back-end logic for the ERP system using ASP.NET Web API against a normalized MS SQL Server schema.
- Engineered dynamic, data-driven UI components with React.js and JavaScript (ES6+), applying MVC design patterns for long-term maintainability.
- Ensured cross-browser compatibility and responsive design using semantic HTML5 and CSS3 best practices.
- Participated in the full software development lifecycle — from requirements gathering and system design through QA, UAT, and production deployment.

## SCAF Solution

Jun 2020 – Dec 2020

### Front-End Developer · Freelance

- Developed and maintained the UI of the Scalf Medical web application using HTML, CSS, JavaScript, and Bootstrap.
- Collaborated with the backend team to integrate front-end and back-end components; managed code with GitHub for version control.

## FREELANCE PROJECTS

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### Dream Labs Smart PP — Hospital Ecosystem (Clinic Management System)

#### Freelance Full Stack Developer | [private-practice-center-client.vercel.app](https://private-practice-center-client.vercel.app)

- Built a multi-tenant clinic management SaaS with React 18, TypeScript, Vite, Node.js, Express, and MongoDB, featuring a full EMR module (13 clinical sections), ICD-10 coding, digital prescriptions, and AI-powered prescription validation.
- Implemented RBAC supporting 5 roles (Doctor, Receptionist, Pharmacist, Clinic Owner, Super Admin) with multi-role switching, and a real-time patient token queue system via Socket.IO.
- Designed RESTful API with JWT access/refresh token auth, rate limiting, and audit logging (2-year TTL); integrated Stripe subscription billing with plan-based feature gating via middleware guards.
- Built pharmacy and inventory management (dispensing queues, batch tracking, expiry alerts, supplier management) and end-to-end billing with PDF receipts and analytics exportable to PDF and Excel.
- Structured as a pnpm/Turborepo monorepo with shared TypeScript and Zod validation across frontend and backend; containerized with Docker Compose and Nginx reverse proxy for production deployment.

### Netexper Pty Ltd — SL Auction

#### Frontend Developer | [slauction.lk](https://slauction.lk)

- Built a high-performance auction platform frontend using Nuxt.js and Tailwind CSS; collaborated with backend developers and UI/UX designers to optimize performance and functionality.

### Escape with Amila Tours — Travel Agency Website

#### Freelance Frontend Developer | [escapewithamila.com](https://escapewithamila.com)

- Designed and developed a fully responsive travel agency website for Escape with Amila Tours, showcasing expert-guided adventure, cultural immersion, and luxury travel experiences worldwide.
- Designed a fully responsive travel website with SEO-optimised metadata, performance tuning, and modern UI/UX; handled end-to-end deployment including domain configuration and production release.

### UdawalawaTaxi — Taxi & Tour Booking Platform

#### Freelance Full Stack Developer | [udawalawataxi.com](https://udawalawataxi.com)

- Built a full-featured taxi and tour booking platform using Next.js and Tailwind CSS for a Sri Lanka-based transport business serving 100+ vehicles.
- Implemented key pages including service listings, transparent rate cards, safari package details, day tour listings (25+ routes), image gallery, and a booking contact form.
- Optimised for SEO with structured metadata, Open Graph tags, and Google-bot directives; integrated WhatsApp and email contact flows for customer conversion; deployed on Vercel with CI/CD pipeline configuration.

## PROJECTS

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### Currency Converter Pro — Chrome Extension

JavaScript · Chrome Extension API · REST API | Published on Chrome Web Store | [lnkd.in/g67BN8-K](https://chrome.google.com/webstore/detail/currency-converter-pro/g67BN8-K)

Hover-based instant currency conversion tool that detects 95+ currencies (symbols & codes) on any webpage, including crypto (BTC, ETH). Privacy-friendly with no tracking, published for travelers, online shoppers, and anyone managing international payments.

### DeadlineBuddy — Chrome Extension

React 18 · Tailwind CSS · Webpack 5 · Chrome Manifest V3 | Published on Chrome Web Store | [lnkd.in/g6eu8gkP](https://chrome.google.com/webstore/detail/deadlinebuddy/g6eu8gkP)

Smart task reminder extension built with React 18 and Chrome Manifest V3, featuring contextual reminders, flexible scheduling (minutes to days), and a sleek dark/light UI. 100% local-first, all data stays private on the user's device.

### SunChat — Real-Time Chat Application

React 19 · Node.js · Express · MongoDB · Socket.IO · Zustand · Tailwind CSS · Vite | [sunchat.onrender.com](https://sunchat.onrender.com)

- Built a React 19 SPA with Vite, Tailwind CSS, DaisyUI, React Router DOM v7, and Zustand for state management; backed by a Node.js/Express REST API with MongoDB/Mongoose for persistent storage and user management.
- Implemented real-time bidirectional messaging via Socket.IO with live online presence tracking; secured with JWT/HTTP-only cookies, bcrypt hashing, and protected route middleware; integrated Cloudinary for profile image storage and deployed on Render.

### BestConverts — Unit Conversion Web App

Next.js 15 · React 19 · TypeScript · Tailwind CSS · Radix UI · Vercel | [bestconverts.com](https://bestconverts.com)

- Built a full-stack unit conversion app supporting 40+ converters across 9 categories including engineering, electricity, heat transfer, fluid dynamics, magnetism, and radiology; architected a custom conversion engine supporting complex formula-based conversions (temperature, thermal conductivity, radiation dose).
- Developed with Next.js 15 App Router, React 19, and TypeScript for type-safe SSR with dynamic routing; implemented SEO features including JSON-LD structured data, sitemap generation, breadcrumb navigation, and robots.txt; designed responsive dark/light UI with Radix UI and integrated Vercel Analytics.

### Kids Stories — Educational Web Platform

Next.js 14 · TypeScript · Tailwind CSS · Cloudflare R2 · Vercel | [kidsstories-bay.vercel.app](https://kidsstories-bay.vercel.app)

SEO-optimized educational platform for children built with Next.js 14 App Router and SSG for fast global delivery. Features interactive PDF story browsing with reading progress tracking, a Countries Explorer, Famous People biographies, and an Animal Kingdom section. Integrated Cloudflare R2 as a CDN for PDF and media assets; mobile-first responsive design with optimized Core Web Vitals.

### My Portfolio

Next.js · Tailwind CSS · Framer Motion · React 18 | [github.com/nuwandm/my-portfolio](https://github.com/nuwandm/my-portfolio) | [nuwandm.github.io/my-portfolio](https://nuwandm.github.io/my-portfolio)

- Built with Next.js 13 SSG and React 18; implemented dark/light mode theming, smooth Framer Motion animations (animated text, magnetic buttons, scroll-triggered transitions), and a custom particle background, animated blob, and tilt card components.
- Applied mobile-first responsive design with Tailwind CSS grid/flexbox, integrated SEO meta tags and optimized image loading, and deployed as a static site on GitHub Pages.

## EDUCATION

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### Sabaragamuwa University of Sri Lanka

2016 – 2022

#### BSc. Honours in Computing & Information Systems

### Kegalu Vidayala

2015

#### G.C.E. Advanced Level

## REFERENCES

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### Isuru Abeywardana

Software Engineer | Bank of Singapore | 2020isuru@gmail.com | +65 8492 1223

### K.P. Sasindu Madushan

Software Engineer | DMS Software Engineering (Pvt) Ltd | kpmnsasindu@gmail.com | +94 71 721 5681